

**Population, sample, parameter, statistic,
sampling bias, simple random sample**

MATH 213

A **population** includes all individuals or objects of interest.

- **Example:** ALL of the words in the Gettysburg address.

A **sample** is the subset of the population about which we have collected data.

- **Example:** The subset of 10 words you selected.

Statistical inference is the process of using data from a sample to gain information about the population.

- **Example:** We tried to use our samples of 10 words to give estimates for the population average word length

A **parameter** is a number (such as a mean, proportion, etc.) that describes a feature of the population.

- **Example:** The average word length among ALL words in the population of words in the Gettysburg Address.

A **statistic** is a number (such as a mean, proportion, etc) that describes a feature of the sample.

- **Example:** The average word length among the 10 words in your sample.

Sampling bias occurs when the method of selected a sample causes the sample to differ from the population in some relevant way. With sampling bias, certain segments of the population can become over-represented, and others under-represented. If sampling bias exists, we cannot trust generalizations from the sample to the population.

- **Example:** Selecting words by eyeballing them resulted in longer words being over-represented in the sample.

In a **simple random sample**, each case from the population has the same chance of being selected, regardless of the other cases chosen for the sample. We will generally select simple random samples using technology.

- **Example:** The 10 words you selected from the Gettysburg address using the random numbers generated by R is a simple random sample.

More complicated random sampling schemes exist and our book describes some of them. We will focus mainly on simple random sampling in our course.